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M T W T F S S

PACKET SNIFFING USING RAW SOCKETS

Aim:

To implement a packet sniffing using Python raw sockets and display the Source IP, destination IP and protocol of captured packets.

CODE:

```
import Socket
Import Struct
Sock = Socket.Socket(3
        Socket.AF_INET,
        Socket.Sock_RAW,
        Socket.IPPROTO_IP)
Print("Packet Sniffer Started...")
print("waiting for packets....\n")
While True:
        Packet, address = Sock.recvFrom(65535)
        ip_header = packet[0:20]
        iph = Struct.unpack("!BBHHHBBH4s4s",
                ip_header)
        Source_ip = socket.inet_ntoa(iph[8])
        destination_ip = Socket.inet_ntoa(iph[9])
        print("-----")
        print("Source IP:", Source_ip)
        print("Destination IP:", destination_ip)
        print("Protocal:", protocal)
        print("packet Size:", len(packet), "bytes")
        print("--------------")
```

SAMPLE OUTPUT:

Packet Sniffer Started...
Waiting For packet.....
Soure IP : 192.168.1.10
Destination IP : 142.250.183.14.
Protocol : 6
Packet Size : 60 bytes.
Source IP: 8.8.8.8.
Destination IP: 192.168.1.10.
Protocol : 1
Packet Size: 84 bytes.

Result:

Thus The Ping Program and Packet Sniffing Program were successfully implemented using Python raw sockets.